AMAZON DEVELOPER HACKATHON 2026 · FIRE TV TRACK

Huddle

Your phones are the Fire TV's camera.

A living-room sports & fitness party game for Fire TV (Vega OS). Up to four players, no extra hardware, and no video ever leaves the phone.

Fire TV · primary | AWS Builder · Bedrock + Polly | Open Source · pose-relay

THE PROBLEM

The biggest screen in the house can't see you.

No camera

Fire TV has no camera, so motion games, fitness form feedback and sports coaching are off the table.

Hardware tax

Today's options are Kinect-style cameras, smart mirrors or $300+ fitness devices, all extra kit for one room.

Tiny screens

Phone fitness apps are single-player and live on a 6-inch screen, which doesn't work for family game night.

THE INSIGHT

Every household already owns 2–4 cameras. They're in your pockets.

1

Scan the QR

Each player scans the code on the TV. It opens a web page, so there's no app install and no account.

2

Prop the phone

MediaPipe Pose runs **on the phone** and tracks 33 body points in real time.

3

Play on the TV

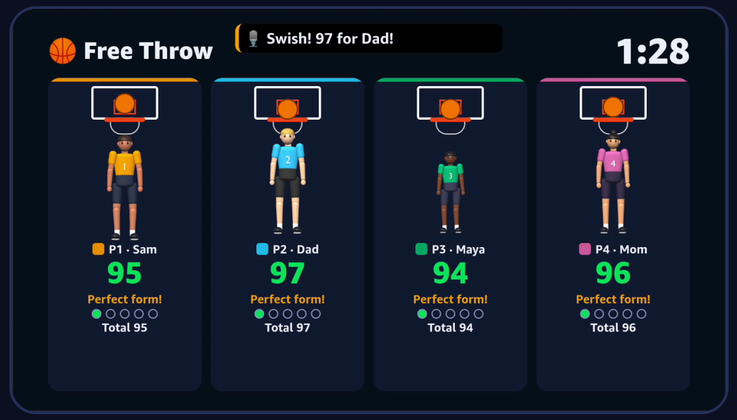
Only keypoints stream to the TV at 15 fps. The TV is the shared arena, with live scoring and an AI coach.

THE PRODUCT

Two games, built for the 10-foot screen



Lobby: QR join, up to 4 players, D-pad menu



Free Throw Form: each shot scored 0–100



Family Boot Camp: rep counting, good / partial

Free Throw Form

Mime a free throw. Each shot is scored on elbow set angle, elbow alignment, knee bend, arm extension and follow-through, with a tip like "*Elbow in!*"

Family Boot Camp

Squats → jumping jacks → high knees, 40 s each. Every threshold is an angle or a body-relative ratio, so a 9-year-old can fairly beat their dad.

TECH IMPLEMENTATION

How it works

📱 Phone page

getUserMedia + MediaPipe PoseLandmarker (WASM/GPU, CPU fallback). EMA smoothing, calibration, **scoring at full frame rate**.

→

☁️ Relay (Node/TS)

Rooms, QR codes, zod validation, 16 KB caps, 40 msg/s rate limits. Built on the open-source **pose-relay** library.

→

📺 Fire TV app

React Native for Vega 0.83. Authority for rounds, timers and totals. D-pad focus, Back/Menu keys, W3C Media audio.

AI coach in the relay

Round summary (numbers only) → **Amazon Bedrock** (Claude via Converse) → ≤ 40-word commentator line → **Amazon Polly** neural MP3 → TV. AWS credentials never reach a client.

Verified

50 unit tests (protocol, scoring, rep counters, rig, rooms, TV logic) and strict TypeScript across the monorepo. Runs on the Vega Virtual Device.

DESIGN

Designed for the couch, not the desk

- **Remote-first:** everything works with D-pad + Select; Back quits a round (with confirm); Menu shows a latency overlay.
- **10-foot UI:** layout authored at 1920×1080 with safe-area margins, high contrast and large focus rings.
- **Fair for all ages:** body-relative thresholds, not pixels or heights.
- **Never silent:** if the coach is slow (> 4 s budget), a template caption shows instead.

Players get characters, not skeletons

Four stylized 3D characters modeled in **Blender**, rendered as shaded body-part sprites and driven by a cutout rig from the 33 landmarks every frame. They run in real time on Vega with plain `Image` views.



AWS BUILDER MINI CHALLENGE

An AI coach that knows your form

SERVICE	HOW HUDDLE USES IT
Amazon Bedrock Converse API	After each round the relay sends scores, ranks and the best and weakest form component (numbers and sanitized names only). Claude returns a family-friendly line that names the winner and gives one concrete tip.
Amazon Polly neural, MP3	Voices the coach line, which the TV plays through the Vega W3C Media `AudioPlayer` with a caption. Boot Camp callouts are synthesized once at startup.
EC2 / Lightsail	Hosts the relay container behind Caddy TLS. App Runner was ruled out because it lacks WebSocket support (logged in our friction log).
IAM	Least privilege: only `bedrock:InvokeModel` and `polly:SynthesizeSpeech`.

PRIVACY & SAFETY

Cameras in the living room, zero video in the cloud.

0

frames of video or images leave the phone. Only 33 × (x, y, visibility) numbers do.

0

accounts. Names live in memory for the room's lifetime only.

4

outbound phone message types (`join`, `pose`, `event`, `ping`). The relay drops everything else.

16 KB

message cap, 40 msg/s rate limit and zod validation on every inbound message.

POTENTIAL IMPACT

Who it's for, and where it goes

Audiences today

- **Families on game night** (parents + kids 8+): an active game everyone plays at once.
- **Home fitness users**: form feedback and motivation without buying hardware.
- **Young basketball players**: objective shooting-form practice.

It works on any Fire TV on Vega OS, using phones people already own.

Roadmap

- **Physical therapy** with clinician-defined exercises and range-of-motion tracking
- **More sports:** golf swing, tennis serve, penalties, boxing combos
- **Alexa+:** "*Alexa, start Boot Camp on Huddle*"
- **Watch-along:** free-throw challenges during live NBA timeouts
- Remote multiplayer between two households' Fire TVs

OPEN SOURCE MINI CHALLENGE

pose-relay: phones as body-pose cameras for any big screen

Server

Transport-agnostic room manager: 4-letter codes, session-token rejoin, heartbeats, rate limits, pluggable validation. About 170 lines with no dependencies.

Phone client

getUserMedia + MediaPipe PoseLandmarker (GPU → CPU fallback) produces mirrored, aspect-corrected keypoints ready for angle math.

Why it matters

Any Fire TV, React Native or web big-screen app can add private multi-player motion sensing in an afternoon. MIT licensed, with a minimal example. Huddle is built on it.

What we learned, and what we'd ask for

Worked well

- About 10 minutes from zero to hello-world on the Vega Virtual Device
- It's just React Native: shared TS packages bundle through Metro
- D-pad focus with no custom code; WebSocket, SVG and audio all available

11 friction-log entries with severity, workaround and suggestion: see FRICTION_LOG.md

Feature requests

- **Companion-device API** with system QR pairing (important)
- **Local WebSocket server** + LAN discovery in Vega apps (important)
- VVD logical resolution aligned with real devices (important)
- `vega device key` / `screenrecord` (nice-to-have)

Turn the living room into the arena.

Code: **github.com/faazkhn/Huddle** · MIT licensed
Library: **github.com/faazkhn/Huddle/tree/main/packages/pose-relay**

Status: runs end-to-end on the Vega Virtual Device with simulated players and a `?sim=1` phone page. The live Bedrock + Polly path is implemented but was not run during the build, which had no AWS credentials; the caption fallback was verified.